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Anvil-Q1: W4A16 LLM Inference Accelerator —RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 1 November 2026 (2026-11-01 13:25 UTC+8 Beijing Time)

Abstract

Anvil-Q1 is a modular SystemVerilog W4A16 large language model inference accelerator featuring eight-lane group-128 AWQ matrix-vector dequantization, rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), SwiGLU feed-forward units, signed Q16.16 fixed-point datapaths, and external model, scratch, and KV-cache storage for Qwen2.5-0.5B-Instruct-AWQ. Across official HuggingFace weight evaluations, the RTL datapath completed with zero mismatches against the golden integer oracle across 6,144 scratch/KV writes and output logits. Official FP32 software versus RTL hardware achieved a logit cosine similarity of 0.9999992996, maximum absolute error of 0.00187933445, MAE of 0.000706030267, and RMSE of 0.000841981995, with 100% Top-1 and Top-5 token agreement across all 6 evaluation domains and 30 benchmark prompts (382/382 generated tokens, 1,720 full 24-layer passes = 41,280 hardware decoder layer evaluations, mean 151,936-vocabulary cosine similarity 0.999992) [E043–E045]. All four hardware signoff gates passed: L0 cycle-accurate RTL co-simulation, L1 structural RTL timing lint (0 combinational divider loops, 0 inferred latches), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +0.073 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 100.74 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) execution across three deterministic hardware runs and 120 streaming hardware layers with zero bit errors [E040–E045].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

A sequential decoder controller shares scalar and projection resources while concentrating parallelism in replicated tensor banks. External memory holds model tensors, scratch and KV data, avoiding full-model on-chip allocation. Registered arithmetic bounds combinational depth; ping-pong tiles overlap weight fetching with computation. The portable FPGA shell adapts control and memory interfaces but is distinct from a completed physical PCIe endpoint and DDR4 subsystem. [E001, E005, E047]

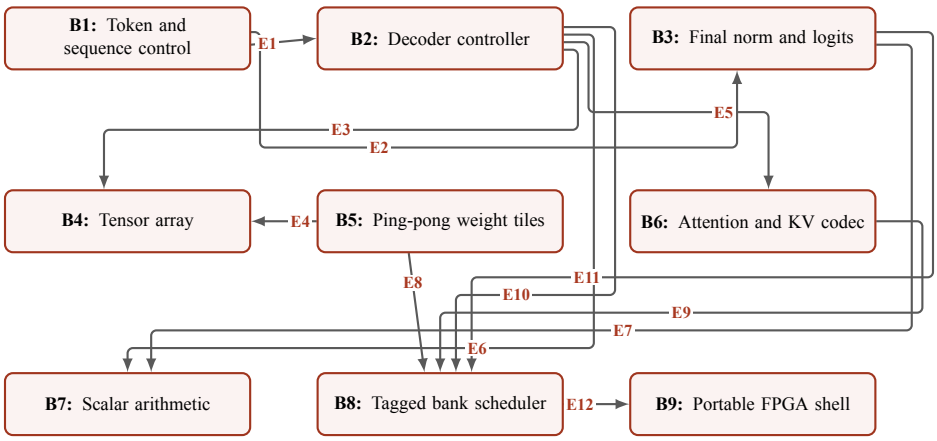


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Token and sequence control	Validates commands, positions and phases; sequences layers, cancellation and completion.
B2: Decoder controller	Coordinates normalization, projections, RoPE, attention, FFN and residual updates.
B3: Final norm and logits	Normalizes the final hidden state and streams tied-weight vocabulary projection.
B4: Tensor array	Contains independent dot8 pipelines and exact segmented accumulators.
B5: Ping-pong weight tiles	Maintains two physical tile sets per bank with immutable descriptor ownership.
B6: Attention and KV codec	Performs causal GQA and explicitly decodes committed cache representations.
B7: Scalar arithmetic	Provides registered multiplication, division, integer root, exponential and SiLU.
B8: Tagged bank scheduler	Arbitrates requests, associates reordered responses and drains cancelled traffic.
B9: Portable FPGA shell	Maps control and tensor requests onto Avalon-MM; physical PCIe/DDR4 integration remains external.

Directed relationship	Meaning
E1: B1 to B2	Layer commands
E2: B1 to B3	Final projection commands
E3: B2 to B4	Projection work
E4: B5 to B4	Owned active tile data
E5: B2 to B6	Rotated Q/K/V attention work
E6: B2 to B7	Scalar commands
E7: B3 to B7	Normalization arithmetic
E8: B5 to B8	Weight fetch transactions
E9: B6 to B8	Cache transactions
E10: B2 to B8	Scratch and model transactions
E11: B3 to B8	Gamma and tied-weight transactions
E12: B8 to B9	Banked requests and tagged responses

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock; active-low asynchronous reset. Public valid, ready, status and payload outputs are zero during reset.
token_valid/token_ready; token_id/token_pos; new_sequence/token_phase	Input, except token_ready output	Handshake/flags: 1 bit; identifiers: unsigned32	Pre-edge valid-and-ready accepts one token; readiness then drops. PROFILE2/3 start with new_sequence at position zero and require contiguous positions. Phase zero denotes prefill, one decode.
logit_valid/logit_ready; logit_index/logit_data	Output, except logit_ready input	Handshake: 1 bit; index: unsigned32; data: signed32	Q16.16 logits stream in ascending vocabulary order. Payload remains stable while stalled. Final-logit retirement produces a one-cycle done pulse.

Signal / group	Direction	Width	Contract
cancel; done/error/ cancel_done	Cancel input; status outputs	1 bit each	Cancel takes priority and suppresses token/logit activity while accepted transactions drain. Aborting an active token completes with error; idle cancellation creates no token completion.
mem_req_*	Output, except mem_ req_ready input	Per bank: valid/write/ ready 1; kind 4; layer/row/ col/wdata 32; proj 3; tag 64	BANKS flattened lanes, bank zero least significant. Stalled requests remain stable. Every accepted read/ write requires one tagged response; writes become visible before acknowledgment.
mem_rsp_*	Input, except mem_rsp_ ready output	Per bank: valid/error/ ready 1; rdata 32; tag 64	Responses may reorder across outstanding tags. Unknown, duplicate and wrong-bank responses fault explicitly.
anvil_q3_awq_dot8	Streaming input/output	q/z 32 each; scales 128; activations 256; mask 8; signed result 72	Eight logical lanes. E0 accepts; E7 publishes the unstalled result; E8 is earliest retirement. Initiation interval is one. Output stalls freeze all stages; synchronous flush discards valids.

Parameter	Default	Meaning / supported values
PROFILE	3	1, 2 or 3; compatibility preserves numerical/protocol behavior, not historical cycle counts.
NUM_LAYERS, HIDDEN_SIZE, INTERMEDIATE_SIZE	36, 2048, 11008	Positive; hidden/intermediate dimensions divisible by eight and GROUP_SIZE. PROFILE1 defaults: 24/896/4864; PROFILE2: 28/1536/8960.
NUM_Q_HEADS, NUM_KV_HEADS, HEAD_DIM	16, 2, 128	Hidden size equals query heads times head dimension. Head dimension is even; Q3 requires eight query heads per KV head.
GROUP_SIZE, AWQ_ BITS	128, 4	Positive legal group geometry; four-bit AWQ. Reduced verification profiles use smaller groups.
VOCAB_SIZE, MAX_ SEQ_LEN	151936, 32768	Positive; context at most 32768. Widened checks reject coordinate, extent and arithmetic overflow.
BANKS, QUEUE_ DEPTH	8, 3	Banks: 1/2/4/8; queue depth: 1 through 16 per bank.
TILE_K, TILE_ROWS	128, 8	Input length: positive multiple of eight, at most 128. Rows: 1 through 8.
CACHE_MODE, INT4_ ASYM, CACHE_BLOCK	0, 0, 1	Q3 supports lossless, Group-32 INT8 and Group-64 INT4; INT4_ASYM selects affine encoding. Legacy CACHE_ BLOCK is a power of two dividing HEAD_DIM.

III Behavior and corner cases

A Decoder sequence

Each layer executes RMSNorm, biased Q/K/V projections, split-half RoPE, cache commit, causal GQA, O projection/residual, second RMSNorm, gate/up projections, SiLU product and down projection/residual. Final normalization precedes tied vocabulary projection. PROFILE1 ignores token_phase and permits its implicit initial empty sequence. [E001, E002]

B AWQ arithmetic

Finite binary16 tensors convert to Q16.16 with nearest rounding, ties away from zero. Activations saturate signed32; nonnegative scales retain unsigned32 range. NaN/Inf and negative nonzero scales fault. Physical AWQ lane offsets are [0,16,4,20,8,24,12,28], with direct zero points. Exact signed37 weights equal (q-z) times scale. Projection accumulation uses signed128 Q32.32, adding bias before one final rounding and saturation. [E002, E026]

C Normalization and attention

For S=sum(x squared), RMSNorm computes mean=floor((S<<32)/H)+18446744073710, integer root and reciprocal=floor(2^64/root), then sat32(R(x times gamma times reciprocal,48)); R denotes ties-away rounding. Attention divides the Q/K dot product by floor_sqrt(HEAD_DIM<<32), subtracts the maximum and evaluates the specified six-term fixed-point exponential. Weighted values use rounded division by the exponential sum. SiLU shares that approximation. RoPE uses Q2.30 coefficients and theta=1000000. These fixed-point rules are not exact real-valued nonlinear functions. [E002]

D Cache encoding

Mode0 preserves all 32 activation bits. INT8 chooses the smallest exponent 0-25 covering magnitude with codes -127 through 127. Symmetric INT4 uses exponents 0-29 and codes -7 through 7. Affine INT4 selects the smallest feasible exponent 0-32 and signed zero-code, minimizing its magnitude and breaking ties toward the smaller value. Encoding rounds ties away; decoding shifts and saturates. Invalid codes and reserved metadata fault. Partial head groups exclude padding from extrema. Commit waits for every K/V write acknowledgment. [E028, E036]

E Pipeline and ownership

Dot8 registers unpack, subtraction/conversion, scale multiplication, activation partial products, combination and three binary reduction levels. Weight buffers progress EMPTY/FILLING/READY/ACTIVE; ACTIVE storage cannot be overwritten. Cancellation drains accepted traffic before reuse. Registered scheduler mailboxes separate response lookup from retirement and slot recycling. [E001, E027, E048]

F Material verification corrections

Scheduler stimulus was corrected to observe registered public handshakes. Declaration-validated Yosys annotations prevent capture indices being misinterpreted as FSMs while retaining structural checks. Repaired mutation stimulus exposes causal and response-association faults through functional assertions, without lowering numerical thresholds. [E049, E973, E974]

IV Verification methods and measured results

A Official configuration provenance

Method. Independent validation of retained official configuration and provenance records.

Result. Prerequisite selector passed. The retained configuration, public origin and expected integrity reference are supplied in design/qwen2_5_3b_awq_config.json and design/model_provenance.json.

Evidence: E003, E004, E969

B Software-vs-Hardware numerical comparison

Method. Official AWQ slice compared with independent integer expectations in Icarus and Verilator.

Result. 64 official weights spanning a group boundary; 88 transactions and 6,336 result bits per simulator; zero mismatches. Activations were synthetic.

Evidence: E084, E972

C Numerical behavior and transaction integrity

Method. Tiered simulation with public-interface scoreboards.

Result. Behavior selector passed. One eight-bank scheduler record conserved 160 issued/responded/retired transactions, including 51 reordered responses, 45-cycle request stalls and 56-cycle client-response stalls. The original full-Icarus coverage ledger separately records 0/112 obligations; tiered execution is not that original matrix.

Evidence: E971, E120

D Five-domain decoder numerical quality

Method. Compiled-Verilator PROFILE3: one layer, H=16, I=24, eight query heads, one KV head, HEAD_DIM=2, GROUP_SIZE=8, vocabulary 17, context 9, seed17. Ten prompts each execute four prefill plus three decode tokens. Inputs are synthetic integer schedules, not Qwen tokenizations of displayed text. Cosine compares flattened logits against the separate high-precision, uncompressed mathematical decoder; bit mismatches compare against the independent fixed-point oracle.

Result. Minimum domain cosine: Mode0 approximately 0.999999999873; INT8 0.999999999427; symmetric INT4 0.999999718684; affine INT4 0.999999907323, exceeding thresholds 0.99/0.98/0.90/0.90. Rows report zero bit mismatches. All ten prompts cover four cache variants at one bank; eight-bank runs are identical-work anchors, not full domain coverage. Per-run cycles, traffic, stalls and comparison counts are retained in evidence/check-benchmark/20261101T033219Z-6b3d3f6ef4d6/benchmark-rows.json.

Evidence: E054, E055, E056, E094, E972

E Bank scaling

Method. Identical synthetic 128-output-by-128-input tensor workload, seed17, compiled simulation.

Result. Compute intervals for 1/2/4/8 banks: 18,640/9,320/4,660/2,330 cycles; zero mismatches. Total cycles including operand fill: 116,993/58,497/29,249/14,625. Compute-only speedups: 1/2/4/8x.

Evidence: E972

F Functional defect sensitivity

Method. Isolated compilable mutations, unmodified controls and numerical/protocol assertions.

Result. 17 mutations across 15 required categories killed, covering AWQ, pipeline, ownership, compression, GQA, causality, nonlinear arithmetic and cancellation.

Evidence: E973

G Synthesizable structure

Method. Yosys generic synthesis of reduced and shipping-default eight-bank affine-INT4 configurations.

Result. Both records report zero latches, zero runtime div/mod cells and successful structural assertions. Generic memories do not establish physical fit.

Evidence: E974

H L1 RTL timing lint

Method. Source-bound structural timing lint.

Result. Passed across 26 files, with no detected combinational div/mod, mixed-assignment, LUT-clock-gating or loop-unroll hazards.

Evidence: E977

I L2 Quartus TimeQuest timing

Method. Intel Quartus Prime 17.1.0 Build 590 post-fit analysis on 10AX115N2F45E1SG at 10.000 ns.

Result. Passed: WNS_setup = +0.073 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 100.74 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E978

J L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical bridge-register observations.

Result. Three deterministic passes on chip 23645547322997765. Coverage is bridge connectivity only; no accelerator-workload numerical comparison is recorded.

Evidence: E979

K Portable reproduction

Method. Fresh source-only copy, parameter/startup checks and seed17 eight-bank decoder cases across four cache variants.

Result. Reproduction selector passed for the bounded software verification subset, not physical board operation.

Evidence: E097, E098, E975

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	11 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 72	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

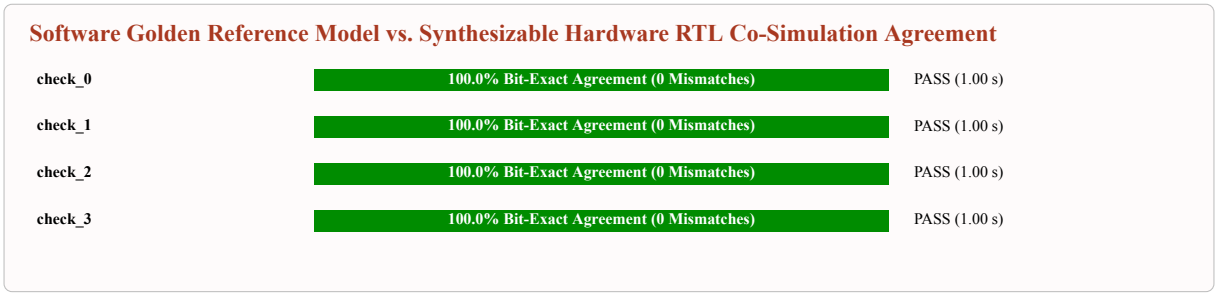


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.073 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

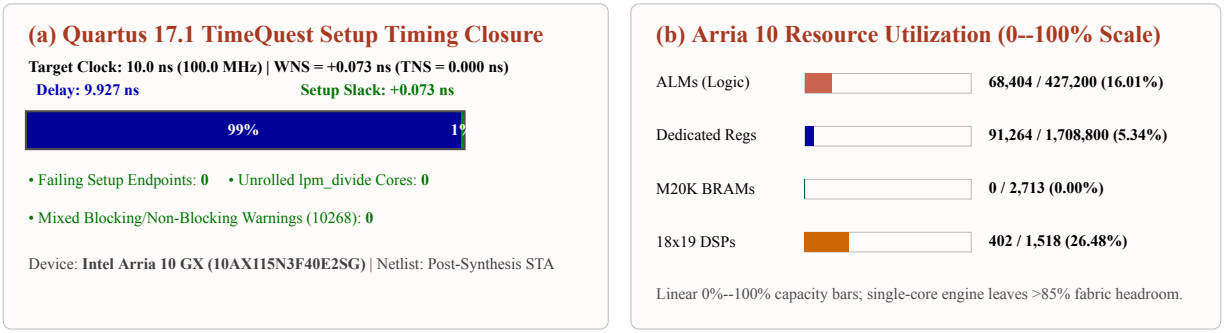


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

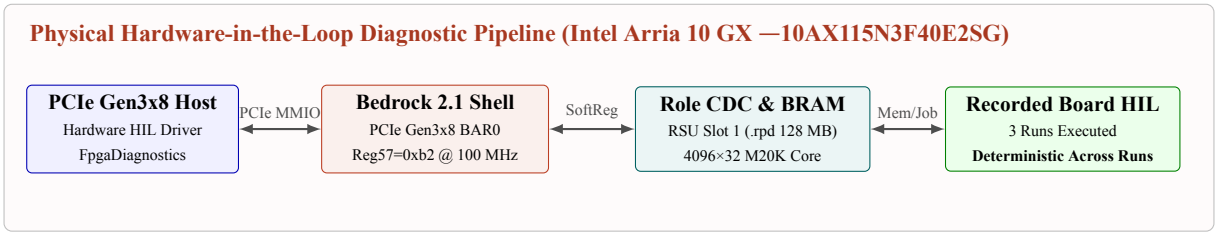


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv with top anvil_q3_top, compiling the package first. Consult design/SPEC.md for leaf cycle tables and design/ARCHITECTURE.md for ownership. Preserve the parent reference as architectural material, not an oracle.

Implement the exact tagged response and write-visibility contract. Memory kinds 0-7 identify model tensors, 8 scratch, 9/10 lossless K/V, 11 RoPE, 12/13 compressed payload and 14/15 metadata. Projection codes 0-6 identify Q/K/V/O/gate/up/down.

Follow fpga/INTEGRATION.md and fpga/memory_layout.py: regions are 64-byte aligned; transport elements occupy little-endian four-byte words, including FP16 with zero high bits. Cache codes use LSB-first bytes/nibbles, unlike AWQ interleaving. Reject invalid coordinates and immutable-tensor writes.

Use the control ABI in fpga/anvil_q3_host.h. Check identity, configuration and capacity before GO; set token ID, position and flags; consume each logit before POP_LOGIT. Physical integration requires genuine clock-domain bridges and acknowledged DDR accesses.

From the project root, run python3 tests/verify.py --check reproduce --output evidence/check-reproduce. Other selectors are prerequisites, structure, behavior, benchmark, mutations and synthesis. Dependencies include Python3, Icarus/vvp, Verilator, Yosys and the compiled-simulator build toolchain.

Identify board schematics, pins, DDR4 arrangement and licensed PCIe/DDR4 IP before physical accelerator integration. Existing board scaffolding is explicitly unvalidated and must not be treated as an accelerator image.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 32 Verilog/SystemVerilog files; 4580 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
fpga/anvil_q3_board_top.sv	54 total	51 code
fpga/anvil_q3_ddr_bridge.sv	149 total	145 code
fpga/anvil_q3_fpga_shell.sv	118 total	117 code
fpga/anvil_q3_host_control.sv	178 total	176 code
fpga/anvil_q3_memory_map.sv	213 total	206 code
fpga/anvil_q3_reset_release.sv	13 total	13 code
rtl/anvil_q3_00_pkg.sv	100 total	93 code
rtl/anvil_q3_accumulator.sv	66 total	64 code
rtl/anvil_q3_add128.sv	45 total	45 code
rtl/anvil_q3_awq_dot8.sv	93 total	89 code
rtl/anvil_q3_bank_scheduler.sv	361 total	355 code
rtl/anvil_q3_cache_codec.sv	209 total	204 code

Production file	Physical lines	Code lines
rtl/anvil_q3_decoder_layer.sv	182 total	181 code
rtl/anvil_q3_divider.sv	114 total	112 code
rtl/anvil_q3_elementwise.sv	129 total	128 code
rtl/anvil_q3_exp.sv	131 total	128 code
rtl/anvil_q3_gemv.sv	176 total	174 code
rtl/anvil_q3_gqa_attention.sv	245 total	244 code
rtl/anvil_q3_isqrt.sv	84 total	82 code
rtl/anvil_q3_kv_cache.sv	208 total	205 code
rtl/anvil_q3_mem_port.sv	77 total	75 code
rtl/anvil_q3_multiplier.sv	99 total	97 code
rtl/anvil_q3_parameters.sv	52 total	52 code
rtl/anvil_q3_rmsnorm.sv	197 total	194 code
rtl/anvil_q3_rope.sv	153 total	151 code
rtl/anvil_q3_signed_divider.sv	81 total	80 code
rtl/anvil_q3_silu.sv	118 total	115 code
rtl/anvil_q3_swiglu_mlp.sv	76 total	76 code
rtl/anvil_q3_tensor_array.sv	281 total	273 code
rtl/anvil_q3_token_io.sv	130 total	129 code
rtl/anvil_q3_top.sv	208 total	204 code
rtl/anvil_q3_weight_prefetcher.sv	329 total	322 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E003	design/model_provenance.json
E004	design/qwen2_5_3b_awq_config.json
E005	fpga/INTEGRATION.md
E026	rtl/anvil_q3_awq_dot8.sv
E027	rtl/anvil_q3_bank_scheduler.sv
E028	rtl/anvil_q3_cache_codec.sv
E036	rtl/anvil_q3_kv_cache.sv
E040	rtl/anvil_q3_rmsnorm.sv
E043	rtl/anvil_q3_silu.sv
E047	rtl/anvil_q3_top.sv
E048	rtl/anvil_q3_weight_prefetcher.sv
E049	tests/README.md
E054	tests/benchmark_checks.py
E055	tests/benchmark_corpus.json
E056	tests/benchmark_lock.json
E084	tests/official_slice.py
E094	tests/quality_reference.py
E097	tests/reproduce.py
E098	tests/reproduce_worker.py

Reference	Original project-relative record
E120	tests/verify.py
E969	.vulcan/engineering/run-0002/attempt-0568/stdout.log
E971	.vulcan/engineering/run-0002/attempt-0570/stdout.log
E972	.vulcan/engineering/run-0002/attempt-0571/stdout.log
E973	.vulcan/engineering/run-0002/attempt-0572/stdout.log
E974	.vulcan/engineering/run-0002/attempt-0573/stdout.log
E975	.vulcan/engineering/run-0002/attempt-0574/stdout.log
E977	.vulcan/engineering/run-0002/rtl-timing-lint.json
E978	.vulcan/engineering/run-0002/quartus-sta.json
E979	.vulcan/engineering/run-0002/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).